

IB MATH AA SL

Lesson 27

Correlation & Linear Regression

Statistics & Probability · 60-minute lesson

Syllabus SL 4.4 · Property of Lynda Badmus Education

A decorative background on the right side of the slide. It features two large, overlapping circles in shades of blue and purple. A large, stylized, light blue letter 'r' is positioned within the upper circle.

Do Now — recall

ON YOUR WHITEBOARD

1. As ice-cream sales rise, do drownings rise too? (data)
2. What does a 'line of best fit' do?
3. If points slope down left→right, is that + or – correlation?

Answers (click to reveal)

1. They correlate (but heat causes both!)
2. models the trend
3. negative

What we're learning today

LEARNING OBJECTIVES

- 1 Describe**
correlation from a scatter diagram.
- 2 Interpret**
the correlation coefficient r .
- 3 Find**
the regression line $y = ax + b$ (GDC).
- 4 Use**
the line to make predictions sensibly.

SUCCESS CRITERIA — I CAN...

- ✓ describe correlation (strength & direction)
- ✓ interpret r between -1 and 1
- ✓ find the regression line on the GDC
- ✓ predict, and know when extrapolation is unsafe

Correlation

r : STRENGTH & DIRECTION

$$-1 \leq r \leq 1$$

r near +1

strong positive

r near -1

strong negative

r near 0

little/no linear correlation

CAREFUL

Correlation $\neq$ causation.

A hidden factor may cause both.

Scatter

always picture the data first.

Interpret r

QUESTION

A data set gives $r = 0.92$.

Describe the correlation.

SOLUTION

r is close to +1 and positive:

→ a strong, positive linear correlation.

As one variable increases, so does the other.

Your turn — describe r

ANSWER IN YOUR BOOK

1. $r = -0.88$. Describe it.
2. $r = 0.10$. Describe it.
3. $r = 0.65$. Describe it.

Answers (click to reveal)

1. strong negative
2. very weak / none
3. moderate positive

Regression & prediction

LINE OF BEST FIT

$$y = ax + b$$

GDC

gives a (gradient) and b (intercept)

Predict

substitute an x to estimate y

Interpolation

inside the data — usually reliable

EXTRAPOLATION

Predicting outside the data range is risky.

The trend may not continue.

Say so

mention reliability in your answer.

Predict with the line

QUESTION

A regression line is $y = 3x + 5$.

Predict y when $x = 4$.

SOLUTION

Substitute $x = 4$ into the line:

$$y = 3(4) + 5$$

Evaluate:

$$y = 17$$

Reliable only if $x = 4$ is within the data range.

Exam-style question

For 10 students, hours studied (x) and test score (y) give $r = 0.86$ and $y = 6.4x + 21$.

- (a) Describe the correlation. [2]
- (b) Estimate the score for a student who studies 5 hours. [2]

MARK SCHEME — reveal after students attempt (tutor only)

- (a) strong positive (linear) correlation (A1)(A1)
- (b) $y = 6.4(5) + 21 = 53$ (M1)(A1)

Common mistakes



Correlation $\neq$ cause

A strong r does not prove one thing causes the other.



r range

r is always between -1 and 1 .



Extrapolation

Predicting far outside the data is unreliable.



Direction

Negative gradient $\rightarrow$ negative correlation.

Mixed practice

QUESTIONS

1. Describe $r = -0.95$.
2. $y = 2x + 7$. Predict y at $x = 6$.
3. Is $r = 0.05$ strong?
4. Is predicting outside the data safe?

Answers (click to reveal)

1. strong negative
2. 19
3. no, essentially none
4. no (extrapolation)

Plenary & exit ticket

KEY FORMULAS

$$-1 \leq r \leq 1$$

$$y = ax + b$$

EXIT TICKET — before you leave

1. $r = 0.97$: describe it.
2. $y = 4x - 1$. Predict y at $x = 3$.
3. Does correlation prove causation?

Answers (click to reveal)

1. strong positive
2. 11
3. no

Self-assess:
• need more help
• nearly there
• confident

Homework

SET ON DR FROST MATHS (auto-marked)

Task:

Correlation & regression.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 28

Probability & Venn diagrams — measuring chance.

$$P(A) = \frac{n(A)}{n(U)}$$